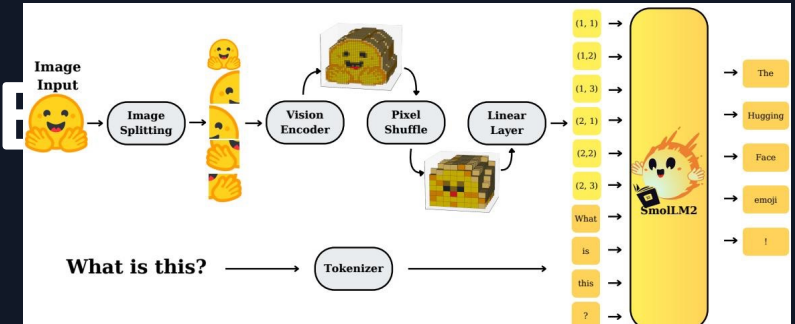


SmolVLM: Redefining Small and Efficient Multimodal Models

Hugging Face · Stanford University

A family of compact vision-language models that achieve competitive performance with dramatically lower memory footprints through strategic architectural optimizations.

Sub-1GB inference · Outperforms 300× larger
Llama-3.1-8B



Large VLMs Are Powerful but Impractical for Edge Deployment

- State-of-the-art VLMs (e.g., Ldefics-80B, Molmo-A1B-7B) require 10–80 GB of GPU RAM, making mobile and edge deployment infeasible.
- Memory bottlenecks dominate inference latency and power consumption on resource-constrained devices.
- Existing 'efficient' open-source models still exceed 20 GB RAM, leaving a gap for truly compact multimodal inference.
- SmolVLM closes this gap: the 256M model runs in <1 GB RAM while outperforming models 300× its size.

Molmo-A1B-7B

27.7 GB RAM

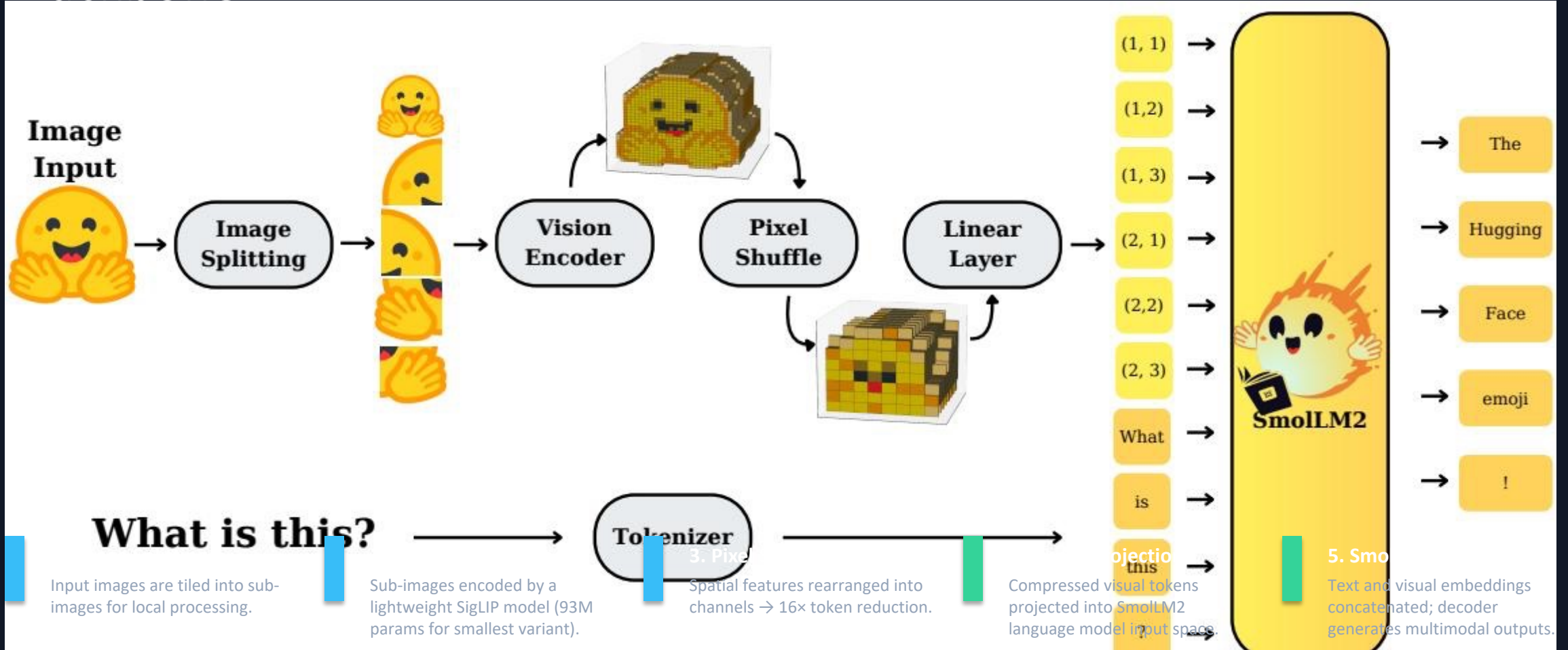
Best Efficient OS baseline

SmolVLM-256M

0.8 GB RAM

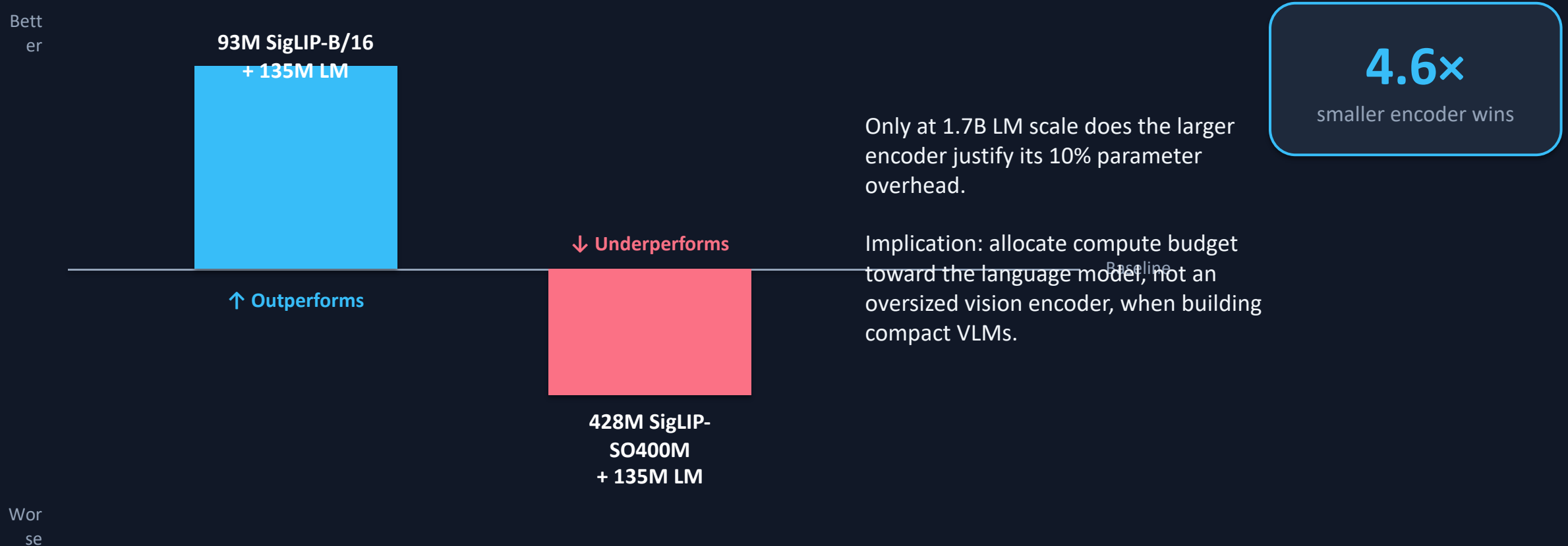
Outperforms Ldefics-80B (80B params)

SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



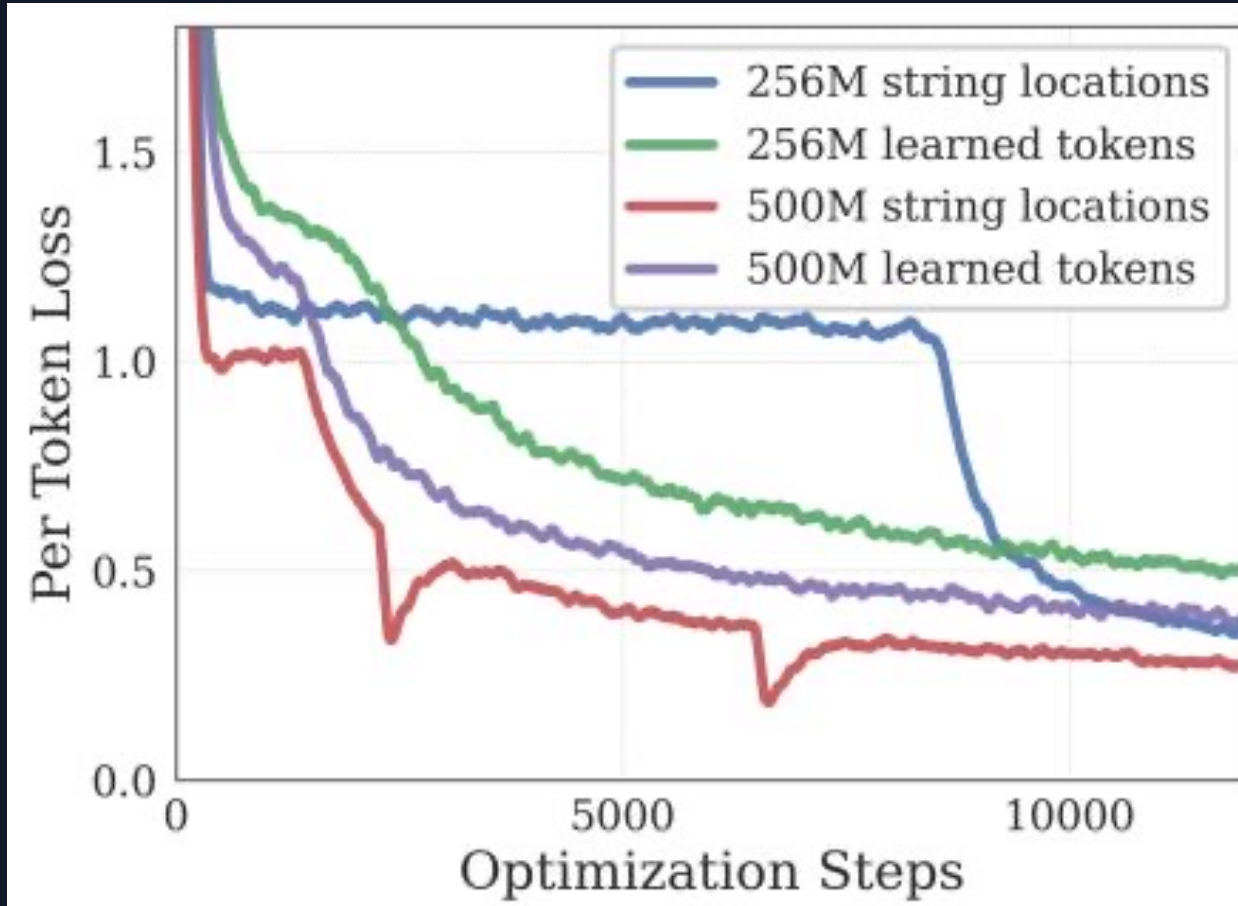
Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1 — Balanced Encoder-LM Compute: compact VLMs benefit from smaller vision encoders.



Aggressive Pixel Shuffle (r=4) Cuts Visual Tokens 16× for Small Models

Finding 3 — Aggressive Token Compression: pixel shuffle rearranges spatial features into channels, reducing tokens by r^2 .



How Pixel Shuffle Works

- Input feature map: $H \times W \times C$
- Rearrange spatial grid into channel depth
- Output: $(H/r) \times (W/r) \times (C \cdot r^2)$
- Result: token count reduced by factor of r^2

r = 2

4× reduction
Preferred by larger models

r = 4

16× reduction
Best for small VLMs

Small VLMs benefit from r=4 because fewer tokens ease attention overhead and improve long-context modeling.

Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Figure 2 — Extended Context Length: performance improves significantly when scaling context from 2K to 16K tokens.

- RoPE base increased from 10,000 → 273,000 to support longer positional encodings without catastrophic attention degradation.
- Fine-tuned on mixed long/short-context data to stabilize training and preserve short-context accuracy.
- SmolVLM-2.2B uses 16K context; 256M and 500M variants use 8K — matching model capacity to context demand.
- Longer context enables higher-resolution image tiling and longer video sequences without truncation.

Context Scaling



Baseline

256M / 500M

RoPE Base: 10K → 273K

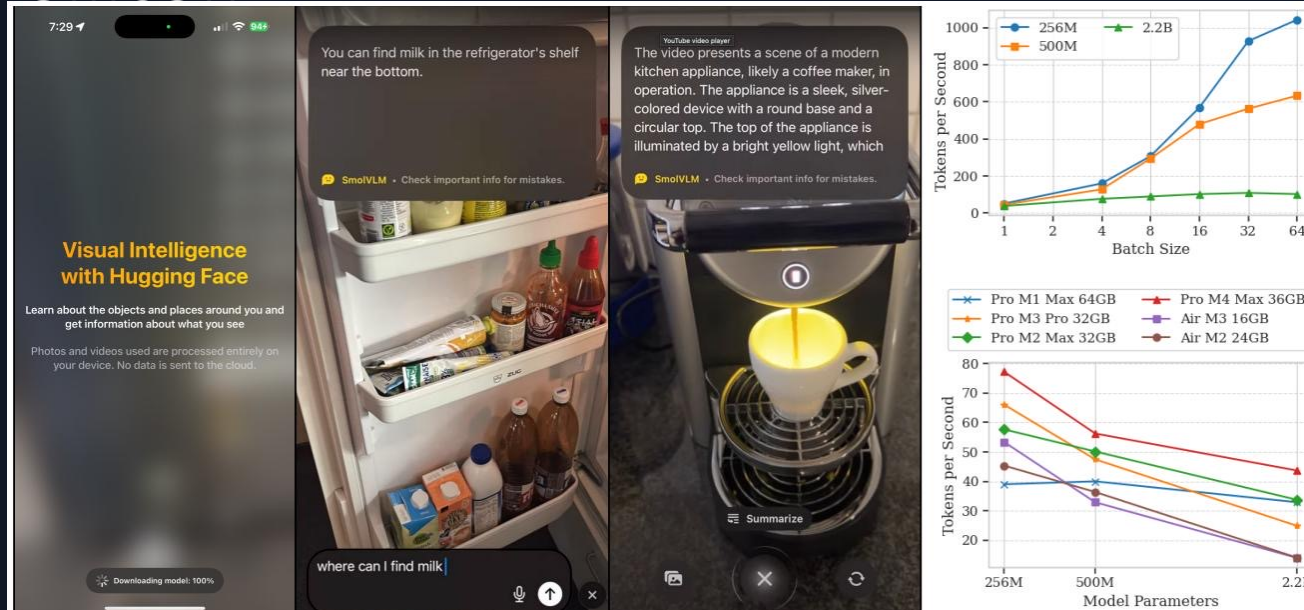
27× increase enables 8× longer context

SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Capability	Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo-A1B-7B)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo-A1B-7B)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo-A1B-7B)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo-A1B-7B)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo-A1B-7B)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo-A1B-7B)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo-A1B-7B)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo-A1B-7B)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo-A1B-7B)
Video	Video-MME	33.7%	42.2%	52.1%	45.0% (InternVL2-2B)
Video	MLVU	40.6%	47.3%	55.2%	48.2% (InternVL2-2B)
Average	All Benchmarks	44.0%	51.0%	59.8%	—

RAM Usage: SmolVLM-256M = 0.8 GB · SmolVLM-500M = 1.2 GB · SmolVLM-2.2B = 4.9 GB · Molmo-A1B-7B = 27.7 GB

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon



- HuggingSnap iOS app processes photos and videos entirely on-device — zero cloud dependency.
- SmolVLM-256M achieves ~80 tokens/second on Pro M1 Max 64GB.
- At batch size 64, the 256M model reaches nearly 1,000 tokens/second on high-end Apple Silicon.
- Even on a MacBook Air M3 16GB, throughput is ~55 tokens/second.
- Throughput scales roughly linearly with batch size for the smallest model, enabling efficient batch serving.

~80 tok/s

Pro M1 Max 64GB

~1,000 tok/s

Batch 64, high-end

~55 tok/s

MacBook Air M3 16GB

Key Takeaways: Design Principles for Efficient Multimodal Models

- Balance encoder-LM compute: smaller vision encoders (93M SigLIP-B/16) outperform oversized ones for compact LMs.
- Extend context length deliberately: scaling RoPE base to 273K and mixing long/short data unlocks 16K context for high-resolution inputs.
- Use aggressive token compression for small models: pixel shuffle at $r=4$ yields 16× fewer tokens, easing attention and enabling long-context modeling.
- Sub-1GB inference is achievable: SmolVLM-256M uses 0.8 GB RAM and surpasses the 300× larger LLaVA-80B — efficiency beats raw scale.
- Fully open-source release: all weights, datasets, code, and the HuggingSnap mobile app are public to accelerate on-device AI research.

Implication for on-device AI: architectural efficiency — not parameter count — is the primary lever for deploying capable multimodal models at the edge.